

Module/Pathway/Taxon Review: Bacterial FabA/FabB Unsaturated-Fatty-Acid Biosynthesis in *Pseudomonas putida* KT2440

Target taxon: *Pseudomonas putida* KT2440 (PSEPK, NCBI:txid160488, proteome UP000000556)
Target bucket: KEGG `ppu01040` "Biosynthesis of unsaturated fatty acids" **Module area:** lipid_cell_envelope_metabolism **Date:** 2026-09-01

1. Executive summary

- The FabA/FabB oxygen-independent unsaturated-fatty-acid (UFA) module is fully **present and essential** in *P. putida* KT2440. All four expected steps are encoded — but by genes that are **absent from the candidate list**.
- The three candidate genes attached to bucket `kegg:ppu01040` (**tesA/PP_2318**, **tesB/PP_4762**, **PP_5331**) are **acyl-CoA thioesterases** (KO K10804/K10805/K10806). They cover **none** of the four FabA/FabB/FabF module steps and are a **mis-scoped / over-propagated** attachment. KEGG map01040 for bacteria displays only the thioesterase-mediated acyl-release reactions, so the bucket harvested the wrong genes.
- The true module genes live on KEGG map `ppu00061` (fatty acid biosynthesis):
- **FabA = PP_4174** (Q88FC4; K01716; EC 4.2.1.59 dehydratase + EC 5.3.3.14 isomerase) → covers **steps 1 & 2** (bifunctional).
- **FabB = PP_4175** (Q88FC3; K00647; EC 2.3.1.41) → covers **step 3**. `fabA` – `fabB` are adjacent and form an **operon**.
- **FabF = PP_1916** (Q88LL4; K09458; EC 2.3.1.179) → covers **step 4**; paralog-ambiguous with **PP_3303**.
- **Direct target-species evidence** (Cronan lab,

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):

the *P. putida* `fabA fabB` operon is **essential** — its inactivation blocks growth unless exogenous UFA is supplied — and the aerobic **DesA** desaturase (KT2440 homolog **PP_0217**, Q88RB6) does **not** rescue UFA synthesis in *P. putida*. This distinguishes *P. putida* from *P. aeruginosa*.

- **Curation bottom line:** Module = **SATISFIED**, but the bucket **needs revision** — re-point it to PP_4174/PP_4175/PP_1916 and detach the three thioesterases.

2. Target-organism pathway definition

Included process (this module): the oxygen-independent branch of bacterial type-II fatty-acid synthesis (FAS-II) that introduces a *cis* double bond during de novo elongation. Specifically: FabA dehydrates (3R)-3-hydroxydecanoyl-ACP to *trans*-2-decenoyl-ACP and isomerizes it to *cis*-3-decenoyl-ACP; FabB commits *cis*-3-decenoyl-ACP to elongation; FabF elongates palmitoleoyl-ACP (16:1Δ9) toward *cis*-vaccenoyl-ACP (18:1Δ11). In *P. putida* the physiological end-products are the membrane-phospholipid acyl chains **16:1Δ9** and **18:1Δ11** (

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Neighboring processes to keep separate: - **KEGG map00061 "Fatty acid biosynthesis"** — the shared FAS-II initiation/reduction/dehydration machinery (AccABCD, FabD, FabG, FabZ, FabI/FabV, FabH). These reactions are shared with saturated-FA synthesis and are *outside* the focused UFA branch, even though FabA/FabB/FabF physically reside on this map. - **KEGG map01040 "Biosynthesis of unsaturated fatty acids"** — as drawn for bacteria, this map is dominated by acyl-CoA **thioesterase** hydrolysis and (eukaryote-style) acyl-CoA elongation/desaturation; it is **not** the FabA/FabB module and is the source of the mis-scoped candidate genes. - **Aerobic desaturation (DesA/DesB)** — an oxygen-*dependent* alternative route to UFAs; present as a homolog (PP_0217) but non-functional as a bypass in *P. putida*. Keep as a distinct module. - **β-oxidation / acyl-CoA activation** (FadD PP_4549/PP_4550) and **cyclopropane-FA synthesis** — downstream/parallel, not part of this module.

Alternate names / database definitions: "anaerobic UFA pathway"; FabA = β-hydroxydecanoyl-ACP dehydratase/isomerase; FabB = KAS I; FabF = KAS II. Note the naming trap: KEGG map01040 ≠ the FabA/FabB module despite the identical English name.

3. Expected step model and satisfiability

#	Module step	Enzyme (activity)	KT2440 gene	KO / EC	Call
1	3-hydroxydecanoyl-ACP dehydration	FabA dehydratase	PP_4174 (fabA)	K01716 / EC 4.2.1.59	covered
2	<i>trans</i> -2→ <i>cis</i> -3-decenoyl-ACP isomerization	FabA isomerase	PP_4174 (fabA, same bifunctional enzyme)	K01716 / EC 5.3.3.14	covered
3	<i>cis</i> -3-decenoyl-ACP committed condensation	FabB / KAS I	PP_4175 (fabB)	K00647 / EC 2.3.1.41	covered
4	palmitoleoyl-ACP → <i>cis</i> -vaccenoyl-ACP elongation	FabF / KAS II	PP_1916 (fabF)	K09458 / EC 2.3.1.179	covered (canonical fabD-fabG-acpP-fabF operon; paralog PP_3303 excluded)

- **Steps encoded by candidate genes:** none. The candidate thioesterases map to no module step.
- **Steps present under a different name:** all four — via PP_4174 (fabA, steps 1+2), PP_4175 (fabB, step 3), PP_1916 (fabF, step 4), none of which is in the candidate list.
- **Steps not expected in this organism:** none are missing. FabF (step 4) is mechanistically an *accessory/elongation* step rather than a UFA-committing step, but it is encoded.

4. Candidate genes and evidence

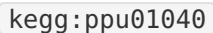
Gene	Locus	UniProt	KEGG KO	Actual function	Relation to module
tesA	PP_2318	Q88KH2 (TrEMBL, PE4 predicted)	K10804	Acyl-CoA thioesterase I / protease I / lysophospholipase L1	Not in module — acyl release/ editing; over- propagated
tesB	PP_4762	Q88DR1 (TrEMBL, PE3)	K10805	Acyl-CoA thioesterase II (EC 3.1.2.20)	Not in module — over-propagated
PP_5331	PP_5331	Q88C52 (TrEMBL, PE3)	K10806	Long-chain acyl-CoA thioester hydrolase	Not in module — over-propagated

Module-relevant genes (not in candidate list, should be added):

Gene	Locus	UniProt	Evidence	Curation notes
fabA	PP_4174	Q88FC4 (Swiss-Prot, PE3 inferred)	Function: direct genetics in <i>P. putida</i> (); sequence: homology	Single K01716; bifunctional dehydratase+isomerase; operon with fabB. High confidence.
fabB	PP_4175	Q88FC3 (TrEMBL, PE3)	Function: direct genetics (operon knockout, ); sequence: homology	Single K00647; essential; internal <i>fabB</i> promoter within <i>fabA</i> CDS. High confidence.
fabF	PP_1916	Q88LL4 (TrEMBL, PE3)	Homology + gene neighborhood (adjacent to fabD PP_1913, fabG PP_1914)	K09458; canonical FabF in core FAS-II cluster. Confident but see paralog.
PP_3303	PP_3303	Q88HQ0 (TrEMBL, PE3)	Homology	Second K09458 KAS-II paralog; needs disambiguation vs PP_1916.

Evidence-type summary: Function of FabA/FabB is supported by **direct experiments in the target species** (genetic operon dissection and essentiality, *P. putida* F1; transfer to KT2440 strong via syntenic PP_4174/PP_4175 operon). *Sequence-level* UniProt evidence for all module genes is homology-based (PE=3/4). The candidate thioesterases have only predicted/homology evidence and no experimental link to UFA synthesis.

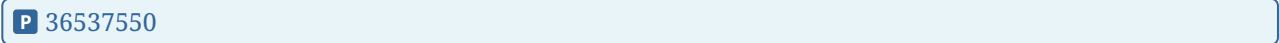
5. Gaps, ambiguities, and likely over-annotations

- 1. Bucket mis-scoping (highest priority):**  attached three thioesterases that satisfy no module step. This is a KEGG-map artifact (map01040 shows thioesterase hydrolysis for bacteria), not a biological gap.
- 2. FabF paralog ambiguity — RESOLVED by gene neighborhood:** PP_1916 vs PP_3303, both K09458/EC 2.3.1.179. **PP_1916** is embedded in the canonical FAS-II core operon

fabD(PP_1913)–fabG(PP_1914)–acpP(PP_1915)–fabF(PP_1916) (all same strand), identifying it as the physiological FabF. **PP_3303** sits between an RND efflux transporter (PP_3302) and a Bcr/CflA multidrug MFS transporter (PP_3304), with no FAS-II genes nearby — a divergent KAS-II paralog unlikely to serve core UFA elongation. Mark PP_1916 **covered (high confidence)**, PP_3303 `candidate_uncertain` (do not use to satisfy the module).

3. **FabH mislabel (over-annotation): PP_4379** (Q88ES4) is annotated "β-ketoacyl-ACP synthase I" in some sources but KEGG assigns **K00648 / EC 2.3.1.180 = FabH (KAS III, initiation)**. It is **not** a FabB/KAS-I and must **not** be used to satisfy the FabB step.

4. **Aerobic desaturase not a bypass:** DesA homolog **PP_0217** (Q88RB6) exists but, unlike in *P. aeruginosa* PAO1, does not supply sufficient UFA when FabA/FabB is inactivated

().

Keep as separate `not_expected_in_target_taxon` for *this* module (it belongs to the aerobic-desaturation module).

5. **Broad/multi-EC mappings:** tesA carries a broad multi-activity annotation (EC 3.1.1.5 / 3.1.2.2 / 3.1.2.14 / protease); such promiscuous entries are prone to over-propagation across lipid buckets.

6. Module and GO-curation recommendations

Per-step status: - Step 1 (FabA dehydratase) → **covered** by PP_4174. - Step 2 (FabA isomerase) → **covered** by PP_4174 (same bifunctional protein). - Step 3 (FabB condensation) → **covered** by PP_4175. - Step 4 (FabF elongation) → **covered (high confidence)** by PP_1916 (canonical fabD-fabG-acpP-fabF operon); paralog PP_3303 marked **candidate_uncertain** and excluded.

Overall module: **SATISFIED** but `module_needs_revision` for the bucket wiring: - **Re-point** the FabA/FabB module to PP_4174, PP_4175, PP_1916 (source map ppu00061). - **Detach** tesA/PP_2318, tesB/PP_4762, PP_5331 from the FabA/FabB module (they may remain in a distinct "acyl-thioesterase / fatty-acyl release" bucket, not a UFA-biosynthesis module). - **Do not** count PP_4379 (FabH) or PP_0217 (DesA) toward this module.

Module-boundary judgment: the *generic* module boundaries are biochemically correct; the error is organism-specific bucket population from KEGG map01040. No change to the generic step model is required, but a curator note should warn that "KEGG map01040 members ≠ FabA/FabB module members" for bacteria.

GO-term suitability (no new GO requests needed): - FabA/PP_4174: GO:0019171 (3-hydroxyacyl-ACP dehydratase) **and** GO:0034017 (trans-2-decenoyl-ACP isomerase) — annotate **both** to capture bifunctionality. - FabB/PP_4175: GO:0004315 (3-oxoacyl-ACP synthase). - FabF/PP_1916: GO:0033817 (β -ketoacyl-ACP synthase II) or GO:0004315. - Parent process: GO:0006636 (unsaturated fatty acid biosynthetic process) for PP_4174/PP_4175.

7. Genes to promote to full fetch-gene review

1. **PP_4174 (fabA)** — primary module gene; add to bucket; annotate dual dehydratase+isomerase activity. *High priority.*
2. **PP_4175 (fabB)** — primary module gene; essential; add to bucket. *High priority.*
3. **PP_1916 (fabF)** — module step-4 gene; confirm as canonical FabF vs PP_3303. *High priority.*
4. **PP_3303** — resolve FabF paralog ambiguity. *Medium.*
5. **PP_4379** — correct FabH vs FabB mislabel. *Medium (annotation fix).*
6. **PP_0217 (desA)** — assign to the aerobic-desaturation module, not here. *Low/context.*
7. **tesA/tesB/PP_5331** — reassign out of the UFA module. *Low (bucket cleanup).*

8. Key references

- Dong H, Wang M, Cronan JE. **Divergent unsaturated fatty acid synthesis in two highly related model pseudomonads.** *mBio / Mol Microbiol* 2023.

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— Direct *P. putida* evidence: fabA-fabB operon (internal fabB promoter), essentiality, DesA not a functional bypass; end-products 16:1 Δ 9 and 18:1 Δ 11.

- Zhu K, Zhang YM, Rock CO. **Transcriptional regulation of membrane lipid homeostasis in *Escherichia coli*.** 2009.

P 19854834

— FabR control of fabA/fabB; defines FabA (β -hydroxydecanoyl-ACP dehydratase/isomerase) and FabB (KAS I) roles (broader-bacterial transfer).

- Marrakchi H, Choi KH, Rock CO. **A new mechanism for anaerobic unsaturated fatty acid formation in *Streptococcus pneumoniae***. 2002.

P 12237320

— Establishes the FabA/FabB anaerobic branch point and the FabM alternative (context for lineage variation).

- KEGG: pathways ppu01040, ppu00061; orthologies K01716, K00647, K09458, K00648, K10804–K10806 (rest.kegg.jp, accessed 2026-09-01).
- UniProt: Q88FC4 (fabA), Q88FC3 (fabB), Q88LL4 (fabF), Q88HQ0 (PP_3303), Q88ES4 (PP_4379/fabH), Q88RB6 (desA/PP_0217).

9. Evidence status and open questions

Supported by direct experiments (target species): FabA/FabB operon structure, essentiality, and the failure of DesA to bypass FabA/FabB loss — all from *P. putida* F1 (

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Transfer to KT2440 is **strong** (syntenic PP_4174–PP_4175 operon with 11-bp gap; near-isogenic strains).

Inferred from homology / pathway DB / gene neighborhood: the KO→step and EC assignments for PP_4174/PP_4175/PP_1916 (KEGG); the FabF assignment to PP_1916 (fabD-fabG-acpP-fabF operon context); the FabH re-identification of PP_4379 (KEGG K00648); the thioesterase identities of the candidate genes (KEGG K10804–K10806). UniProt sequence-level evidence for every module gene is PE=3/4 (no direct protein-level proof in KT2440 specifically).

Open questions / experiments to resolve gaps: 1. *KT2440-specific confirmation*: has the fabA-fabB knockout / UFA-auxotrophy phenotype been reproduced in KT2440 itself (vs F1)? A defined KT2440 ΔfabAB with UFA rescue would upgrade transfer from "strong inference" to "direct." 2. *FabF vs FabB division of labour*: does PP_1916 (FabF) elongate palmitoleoyl-ACP to *cis*-vaccenoyl-ACP and show the temperature-dependent 18:1Δ11 regulation seen in *E. coli*? Lipidomics across growth temperatures would confirm step 4. 3. *PP_3303 function*: what does the efflux-associated KAS-II paralog do? Its neighborhood suggests a role outside membrane-phospholipid UFA synthesis (e.g., specialized acyl chain modification) — a targeted deletion + lipidomics would clarify whether it ever contributes to the module. 4. *DesA (PP_0217)*

conditions: are there conditions (e.g., specific carbon sources, stress) under which the KT2440 DesA becomes functionally relevant? Cronan 2023 showed it is non-functional as a bypass, but conditional relevance is untested.

10. Limitations

- No experimental data files were provided; all conclusions rest on public databases (KEGG, UniProt) and literature.
- The strongest functional evidence is from *P. putida* F1, not KT2440 directly (though loci are syntenic).
- No sequence alignment/BLAST was run in-session; paralog calls rest on KO identity plus gene neighborhood, which is strong but not a substitute for phylogenetic analysis.

Confidence & species-transfer statement

FabA/FabB presence, essentiality, and operon structure are **directly demonstrated in *P. putida*** (F1; strong transfer to KT2440 via identical syntenic loci). Enzyme-to-step assignments rest on KEGG KO/EC plus gene neighborhood (strong for FabA/FabB, good-but-paralog-flagged for FabF). Sequence-level UniProt evidence is homology-based for all module members. The over-annotation and mis-scoping conclusions are supported directly by KEGG map membership.